

SANGMIN LEE

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ABOUT ME

I am a Ph.D. candidate in Electrical and Electronic Engineering at Yonsei University, advised by Prof. Hong-Goo Kang in the DSP & AI Lab.

My research focuses on multilingual speech and natural language processing, with a particular interest in building scalable and inclusive speech AI systems. I aim to develop generalizable models that can understand and generate both speech and text across diverse languages, including low-resource and unseen languages, to extend the benefits of technology to underrepresented and low-resource communities.

RESEARCH INTEREST

Massively Multilingual Speech and Natural Language Processing: Automatic speech recognition (ASR), text-to-speech synthesis (TTS), speech translation, and low-resource language technologies.

Large Speech-Language Models: Unified modeling of speech and text for multilingual spoken language understanding and generation.

Speech Tokenization and Representation Learning: Learning discrete and continuous speech representations for scalable multimodal systems.

Multimodal AI: Integrating speech, language, and other modalities to enable richer contextual reasoning and interaction.

EDUCATION

Yonsei University

Seoul, South Korea

Ph.D. in Electrical and Electronics Engineering

03/2024 – 02/2028 (expected)

- Advised by Prof. Hong-Goo Kang, DSP & AI Lab
- GPA: 3.94 / 4.3 (equivalent to 4.12 / 4.5)

Korea University

Seoul, South Korea

B.S. in Computer Science and Engineering

03/2018 – 02/2024

- GPA: 3.76 / 4.5

WORK EXPERIENCE

DSP & AI Lab, Yonsei University

Seoul, South Korea

Integrated M.S. & Ph.D. Student

03/2024 – Present

DSP & AI Lab, Yonsei University

Seoul, South Korea

Undergraduate Research Intern

01/2024 – 02/2024

CV Lab, Korea University

Seoul, South Korea

Undergraduate Research Intern

08/2023 – 12/2023

Neoforce Incorporation

Daejeon, South Korea

Full-Stack Developer

10/2021 – 02/2022

Republic of Korea Army

Incheon, South Korea

Military Service

04/2020 – 10/2021

AWARDS & HONORS

AI Research Computing Support Initiative

Computing Resources

Korea Electronics Technology Institute / Samsung SDS (South Korea)

06/2026

- Student lead for the project “Efficient Omnilingual Speech Large Language Model (SLLM) via Orthography Unification”, awarded access to 16× NVIDIA H100 GPUs (2 nodes).

National Research Foundation Research Grant	Research Funding
National Research Foundation (South Korea)	03/2026
Student lead in preparing the proposal for the 3-year project “Efficient Omnilingual Speech Large Language Model (SLLM) via Orthography Unification.” (Granted for Core Research Type A)	
NIA Director’s Award, 2023 Korean AI Contest Track 1	Awards
Ministry of Science and ICT (South Korea)	10/2023
Collaborated with 4 teammates to train an automatic speech recognition model for dialects, the elderly, and children.	

PUBLICATION

Conference Papers

- C5. MoLGE: Mixture of Language Group Experts for Efficient Scaling of Massively Multilingual Speech Recognition**
Sangmin Lee, Woojin Chung, Woongjib Choi, and Hong-Goo Kang— *COLM 2026 (Poster)*
- C4. UR-BERT: Scaling Text Encoders for Massively Multilingual TTS Through Universal Romanization and Speech Token Prediction**
Sangmin Lee, Eekgyun Ahn, Woongjib Choi, and Hong-Goo Kang— *Interspeech 2026 (Poster)*
- C3. UniverSR: Unified and Versatile Audio Super-Resolution via Vocoder-Free Flow Matching**
Woongjib Choi, Sangmin Lee, Hyeongseob Lim, and Hong-Goo Kang— *ICASSP 2026 (Poster)*
- C2. UniCoM: A Universal Code-Switching Speech Generator**
Sangmin Lee, Woojin Chung, Seyun Um, and Hong-Goo Kang— *Findings of EMNLP 2025 (Poster)*
- C1. LAMA-UT: Language Agnostic Multilingual ASR Through Orthography Unification and Language-Specific Transliteration**
Sangmin Lee, Woojin Chung, and Hong-Goo Kang— *AAAI 2025 (Oral)*

Preprints

- P3. AEGIS: Awareness-Enhanced Guidance for Iterative Safeguard**
Kyungwon Park, Sangmin Lee, Heejae Chon, and Hyungu Kang— *arXiv:2607.17713*
- P2. SAGE-LD: Towards Scalable and Generalizable End-to-End Language Diarization via Simulated Data Augmentation**
Sangmin Lee, Woongjib Choi, Jihyun Kim, and Hong-Goo Kang— *arXiv:2510.00582*
- P1. Talk3D: High-fidelity Talking Portrait Synthesis via Personalized 3D Generative Prior**
Jaehoon Ko, Kyusun Cho, Joungbin Lee, Heeji Yoon, Sangmin Lee, Sangjun Ahn, and Seungryong Kim— *arXiv:2403.20153*

LANGUAGES

Korean	■ ■ ■ ■ ■
English	■ ■ ■ ■ ■

TECHNICAL SKILLS

Python	■ ■ ■ ■ ■
PyTorch	■ ■ ■ ■ ■
CUDA	■ ■ ■ ■ ■
Linux	■ ■ ■ ■ ■
C/C++	■ ■ ■ ■ ■